

DATA SCIENCE RESOURCES

Python & Pandas Projects

Project 1: Weather Data Analysis

Problem Statement: Weather patterns impact everything from agriculture to logistics. Analyzing weather trends helps in forecasting and planning across sectors.

Objective: Analyze weather data to identify temperature patterns, precipitation trends, seasonal changes, and potential anomalies.

Dataset Source

- [Global Historical Weather Data \(NOAA\)](#)
- [Weather Dataset from Kaggle](#)

Key Analysis Areas

- Daily/Monthly average temperature trends
- Heatmaps of temperature and rainfall by region
- Extreme weather event detection
- Year-over-year comparison

Tools Used: Python (Pandas, Seaborn, Matplotlib), Tableau, Power BI

Expected Outcome: Visual and statistical insights into climate patterns that can inform decision-making for agriculture, energy, and public safety.

Project 2: COVID-19 Data Tracker

Problem Statement: Tracking the spread of COVID-19 helps in evaluating the effectiveness of public health policies and predicting future outbreaks.

Objective: Create a dynamic dashboard to monitor COVID-19 cases, deaths, and recoveries over time and across regions.

Dataset Source

- [Johns Hopkins CSSE COVID-19 Data](#)
- [Our World in Data COVID-19 Dataset](#)

Key Analysis Areas

- Daily case and death trends
- Vaccination progress
- Country-wise and state-wise heatmaps
- Case fatality rate and recovery trends

Tools Used: Power BI, Tableau, Python (Plotly, Dash), Excel

Expected Outcome: An interactive dashboard for public use or internal reporting, highlighting trends and hotspot zones.

Project 3: Global Terrorism Dataset Analysis

Problem Statement: Understanding global terrorism patterns helps governments and researchers identify high-risk regions and develop prevention strategies.

Objective: Explore terrorism data to analyze attack frequency, types, locations, casualties, and group activities over time.

Dataset Source

- [Global Terrorism Database \(GTD\)](#)
- [Available on Kaggle as a structured CSV dataset](#)

Key Analysis Areas

- Top countries and regions by number of attacks
- Attack type distribution (bombing, armed assault, etc.)
- Year-wise casualties and fatalities
- Active terrorist groups and their target types

Tools Used: Python (Pandas, Seaborn), Tableau, Power BI, GeoPandas for maps

Expected Outcome: An analytical report or dashboard that helps visualize global threat trends, target types, and geographical hotspots.

Project 4: FIFA World Cup Data Exploration

Problem Statement: Football analysts and fans are always curious about historical team and player performances, winning strategies, and match statistics.

Objective: Explore historical FIFA World Cup data to extract insights on top scorers, team performance, goal trends, and match outcomes.

Dataset Source

- [FIFA World Cup Data \(Kaggle\)](#)
- [FIFA official statistics database](#)

Key Analysis Areas

- Most goals by team and player
- Match outcome breakdown (win/loss/draw)
- Country performance over the years
- Goal distribution by stage (group vs knockout)

Tools Used :Python, Tableau, Excel, Power BI

Expected Outcome: An engaging analytical dashboard for fans, journalists, or sports data scientists to analyze team strengths and historical records.

Project 5: Olympics Dataset Insights

Problem Statement: The Olympics host thousands of athletes, yet countries and athletes differ significantly in performance and participation trends.

Objective: Explore historical Olympics data to discover medal trends, country-wise dominance, athlete participation, and sport popularity.

Dataset Source: [Olympics Dataset \(Kaggle\)](#)

Key Analysis Areas

- Total medals by country

- Gender-wise participation trends
- Dominant sports by nation
- Medal distribution over time

Tools Used: Power BI, Tableau, Python (Seaborn, Matplotlib)

Expected Outcome: A full analytical report or interactive dashboard that showcases global sports trends, equity in participation, and medal dominance.
